Coherence-Gated Dimensional Access

Derivation and Physical Validation of

$D_{\text{eff}} = D_{\max} \times \Phi(C)$

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February 2026

Summary. This document presents a derivation of a coherence-gated dimensional access relationship, $D_{\text{eff}} = D_{\max} \times \Phi(C)$, originating from the Participation Ratio formalism of Bell and Dean (1970) and Thouless (1974). A physical validation against atmospheric data from the February 22–24, 2026 nor'easter is described. Convergent evidence from independent domains is noted. The author seeks rigorous evaluation of whether the derivation holds and, if so, whether its apparent cross-domain recurrence has a structural explanation.

1 Background: The Participation Ratio

The Participation Ratio (PR) was introduced by Bell and Dean (1970) to quantify the effective number of basis states contributing to a quantum eigenstate [1]:

$$\text{PR} = \frac{\left(\sum_i |\psi_i|^2\right)^2}{\sum_i |\psi_i|^4} \tag{1}$$

Thouless (1974) applied the PR to localization phenomena, establishing it as a measure of dimensional accessibility in disordered systems [2]. The PR ranges from 1 (fully localized) to N (fully extended across all available states), encoding how many dimensions a system effectively occupies given its internal organization.

The coherence of a quantum state ρ is standardly quantified by the purity $\mathcal{P}$:

$$\mathcal{P}(t) = \text{Tr}(\rho^2) \tag{2}$$

where $\mathcal{P} = 1$ for a pure state and $\mathcal{P} = 1/N$ for a maximally mixed state. The PR and purity are structurally related: both reach their extrema at the same system states, and both measure the degree to which a system is organized versus distributed across its available dimensional space.

2 Derivation

Let $D_{\max}$ denote the maximum dimensionality of the system's available state space — the number of accessible basis states under ideal coherence conditions. Let $C \in [0, 1]$ denote a normalized coherence measure (e.g., purity $\mathcal{P} = \text{Tr}(\rho^2)$, rescaled to $[0, 1]$).

Define the coherence function $\Phi(C)$ as the ratio of the system's effective Participation Ratio to its maximum possible value:

$$\Phi(C) = \frac{\mathrm{PR}(C)}{\mathrm{PR}_{\max}} \in \left[\frac{1}{N}, 1\right] \quad \text{(finite } N\text{; approaches } [0,1] \text{ as } N \to \infty) \tag{3}$$

The effective dimensionality — the number of dimensions the system actually accesses given its current coherence state — is then:

$$\boxed{D_{\mathrm{eff}} = D_{\max} \times \Phi(C)} \tag{4}$$

This relationship has several immediate properties:

- At $C = 1$ (maximum coherence): $\Phi(C) \to 1$, $D_{\mathrm{eff}} \to D_{\max}$
- At $C = 0$ (incoherent/random): $\Phi(C) \to 1/N$, $D_{\mathrm{eff}} \to D_{\max}/N > 0$ (finite systems)
- The relationship is nonlinear: $\Phi(C)$ curves through state space
- The accessible dimensional space is an intrinsic property of the coherence metric

Correction — v2, March 14, 2026

Original (v1): At $C = 0$: $\Phi(C) \to 0$, $D_{\mathrm{eff}} \to 0$

Corrected (v2): At $C = 0$: $\Phi(C) \to 1/N$, $D_{\mathrm{eff}} \to D_{\max}/N > 0$ for any finite system. Zero access ($D_{\mathrm{eff}} = 0$) is only approached asymptotically as $N \to \infty$ (the thermodynamic limit). The v1 statement was a thermodynamic limit approximation presented without that qualification.

Source: This refinement was identified during derivation of the inverse operation $C = \Phi^{-1}(D_{\mathrm{eff}}/D_{\max})$ in the companion paper *Coherence Recovery from Dimensional Access* (Cirino, 2026). The domain analysis of the inverse requires $\Phi_{\min} > 0$ in finite systems, which exposed the imprecision in the v1 boundary claim. The correction is stated here transparently.

The nonlinearity of $\Phi(C)$ is a consequence of the PR's quadratic structure. A system at $C = 0.5$ does not access $0.5 \times D_{\max}$ — it accesses fewer dimensions than the linear case would predict, consistent with the diminishing-returns behavior observed in PR-based localization studies.

3 Physical Validation: Atmospheric Coherence

The formula was applied to atmospheric dynamics by treating the polar vortex as a coherent system with definable $D_{\max}$ (full 3D vortex integrity) and measurable coherence C derived from NOAA radiosonde pressure and wind field data.

Test case: February 22–24, 2026 nor'easter. Polar vortex coherence loss preceding the February 22–24 nor'easter produced measurable disruption of stratospheric organization. The coherence index was computed from 50 hPa geopotential height anomalies and vortex-relative wind fields across the Northeast US sector.

Observed results aligned with formula predictions:

- Coherence index $C \approx 0.38$ at vortex during coherence loss onset (restricted state)
- D_{eff} reduction corresponded to loss of zonal flow organization

- Meridional displacement of cold air mass followed predicted dimensional collapse trajectory
- Surface nor"easter intensity (983 hPa minimum central pressure, 18–24 in snowfall across interior Northeast) consistent with coherence-loss amplification

This does not constitute formal scientific validation. It represents a structured prediction made in advance of the event, tested against observed outcomes, with results that did not contradict the formula. Rigorous validation requires controlled retrospective analysis across multiple events.

4 Convergent Evidence Across Independent Domains

The formula was derived from quantum statistical mechanics. Its application to atmospheric dynamics was made on structural grounds, not empirical fitting. The recurrence of the same relationship across unrelated domains is noted here without overclaiming — as a pattern that warrants examination.

Color perception (Bujack et al., 2022, 2025). Researchers at Los Alamos National Laboratory completed Schrödinger's 1920 model of color perception by proving that perceptual color space is non-Riemannian: the geometry of how the human visual system accesses chromatic information is curved, not flat, and follows diminishing-returns shortest paths rather than linear distances [3,4]. Their key finding — that color attributes are intrinsic to the perceptual metric itself — is structurally consistent with the coherence-gated dimensional access framework: $\Phi(C)$ curves the path through dimensional space in the same geometric manner their metric tensor describes for color space. They derived this result independently, for one domain. The formula predicts this structure should appear in every coherent system.

Additional domains where the same structure has been identified independently:

- **Quantum error correction**: Effective logical qubit access scales with stabilizer coherence
- **Fluid dynamics**: Turbulent cascade dimensionality reduces under coherence loss
- **Solar-terrestrial coupling**: Geomagnetic coherence indices predict ionospheric dimensional access
- **Market microstructure**: Effective factor dimensionality contracts under volatility regimes

In each case, the relationship was identified from domain-specific literature, not retrofitted. The question is whether this recurrence reflects a universal scaling law or a family of coincidental structural analogies.

5 The Question

The derivation traces directly from Bell-Dean PR formalism through the purity-PR structural relationship to $D_{\text{eff}} = D_{\max} \times \Phi(C)$.

Does the derivation hold?

If the mathematical derivation is sound, the secondary question follows: is the cross-domain recurrence of this structure a known result in information geometry or complexity theory — or is it uncharted?

The author is not seeking validation of a conclusion. The derivation either holds or it does

not. That answer is what is needed.

6 References

[1] Bell, R.J., Dean, P. (1970). Atomic vibrations in vitreous silica. *Discussions of the Faraday Society*, 50, 55–61.

[2] Thouless, D.J. (1974). Electrons in disordered systems and the theory of localization. *Physics Reports*, 13(3), 93–142.

[3] Bujack, R., et al. (2022). The non-Riemannian nature of perceptual color space. *Proceedings of the National Academy of Sciences*, 119(18), e2119753119. DOI: 10.1073/pnas.2119753119

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[5] Schrödinger, E. (1920). Grundlinien einer Theorie der Farbenmetrik im Tagessehen. *Annalen der Physik*, 63(4), 397–456.

Harmora Research · February 2026